

Will Cygan

SOFTWARE ENGINEER WORKING ON THE PURCHASING PLATFORM AT LINKEDIN

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Work Experience

Senior Software Engineer

Chicago, IL

LINKEDIN

March 2024 - Present

- Architected a high-performance alerting system using Kafka, Samza, and Venice, processing 50,000+ QPS to provide real-time invoice tracking and critical financial alerts for LinkedIn's Global Alerts platform.
- Reclaimed \$2M+ in annualized revenue by preventing involuntary churn through the Global Alerts system, directly impacting LinkedIn's bottom-line and enhancing customer retention.
- Pioneered a scalable Oracle-to-MySQL migration framework, empowering a dozen engineering teams to seamlessly onboard to MySQL, significantly accelerating organizational database modernization efforts.
- Orchestrated the design and execution of a zero-downtime migration for 2 key services from Oracle to MySQL; ensured strong data consistency throughout the transition by leveraging a Couchbase-backed pinning strategy for entity routing.
- Spearheaded the migration of 4 critical services from Rest.li to gRPC, enhancing system interoperability and boosting developer productivity through standardized tooling and comprehensive documentation.
- Optimized the Global Alerts system by implementing an Airflow-based offline workflow to proactively identify and purge stale cache records, successfully cutting downstream service traffic by 50%.
- Led JVM optimization initiatives, employing A/B testing methodologies to elevate JVM health from a baseline of 30-80% to a consistent 99.9%+ across critical, underperforming production services.

Software Engineer

San Francisco, CA

LINKEDIN

Feb 2022 - March 2024

- Developed and scaled backend microservices for "Videos You Might Be Interested In" (VYMBII), a LinkedIn Learning discovery feature on the main feed, handling 3,000 QPS.
- Engineered the backend data infrastructure powering personalized video course recommendations within a TikTok-style carousel, achieving a 10%+ uplift in member engagement with learning content.
- Engineered robust offline data pipelines using Spark and HDFS for "Learning Alerts," a system classifying 10M+ users weekly to deliver targeted course recommendations based on job-seeking signals.

Projects

Anton (Kubernetes Homelab)

🐙 wcygan/anton

CLUSTER OPERATOR / TECHNICIAN

December 2024 - Present

- Constructed a 3-node Kubernetes control plane on Talos Linux, achieving a fault-tolerant, highly available cluster.
- Implemented zero-trust remote access via Cloudflare Tunnels and Tailscale, streamlining remote cluster management
- Deployed data infrastructure like TiDB, RedPanda, DragonflyDB, and ScyllaDB for experimentation & learning

tokio-utils Rust Library

🐙 wcygan/tokio-utils

MAINTAINER

March 2023 - Present

- Crafted a library for Asynchronous Rust which supports patterns like Rate Limiting, Object Pooling, and Graceful Shutdown
- Engineered a web crawler atop this library with rate limiting to ensure continuous data capture of internet graph data (similar to Google's PageRank) without triggering rate limits or IP blacklisting

Skills

Languages Java, Rust, Go, SQL, Python, Scala, Typescript, Bash

Technologies Temporal Workflows, gRPC, Protocol Buffers, Docker, Kubernetes, Talos Linux

Data Systems MySQL, Kafka, Flink, Beam, Spark, Hadoop, Hive, Trino, Kusto, Redis

Education

University of Illinois at Chicago

Chicago, IL

B.S. IN COMPUTER SCIENCE

January 2018 - December 2021